

Mentor–Mentee AI Guidance Platform – Knowledge Base (For RAG)

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1. Project Overview

This document is a structured knowledge base for building a Retrieval-Augmented Generation (RAG) system for a Mentor–Mentee AI Guidance Platform. It includes product scope, features, user flows, system architecture, data schema, APIs, AI workflows, safety, evaluation, and FAQs. The RAG system should retrieve relevant sections from this document to answer user queries accurately.

2. Product Goals

- Enable mentees to discover mentors and book sessions. - Provide AI guidance when mentors are unavailable. - Support real-time sessions with WebRTC. - Provide feedback using speech-to-text and sentiment analysis. - Improve outcomes via learning roadmaps and reminders.

3. Core Features

3.1 Mentor Discovery & Booking - Search mentors by skill, availability, rating. - Book sessions via chatbot using tool calling. 3.2 AI Guidance Agent - AI agent answers queries when mentor is offline. - Uses RAG over platform knowledge + user context. 3.3 Real-time Sessions (WebRTC) - Video/audio calls. - Session notes and summaries. 3.4 Whisper-based Feedback - Transcribe session audio. - Provide communication feedback to mentor and mentee. 3.5 Sentiment Check (Pre-session) - Detect energetic/nervous/neutral mood. 3.6 AI Reminders - Notify mentor/mentee about upcoming sessions.

4. High-Level Architecture

Frontend: React (or Next.js) Backend: Node.js + Express Auth: JWT DB: MongoDB Realtime: WebRTC + Socket.io AI: LLM API + Whisper STT Vector DB: Chroma/FAISS/Pinecone RAG Pipeline: Embed docs → store → retrieve → generate

5. RAG Pipeline Design

5.1 Data Sources - Product documentation - Mentor profiles - Session policies - FAQs 5.2 Ingestion - Chunk documents (300–800 tokens) - Generate embeddings - Store in vector DB 5.3 Retrieval - Top-k semantic search - Re-rank if needed 5.4 Generation - Inject retrieved context into prompt - Return concise, grounded answers

6. Tool Calling (Examples)

Tools: - book_session(mentor_id, date, time) - cancel_session(session_id) - list_available_mentors(skill) - send_reminder(user_id, session_id) Agent Policy: - Ask clarifying questions when required. - Execute tools only with explicit user intent.

7. Data Models (Simplified)

User { id, role: mentor|mentee, name, skills, availability } Session { id, mentor_id, mentee_id, time, status } Feedback { session_id, transcript, sentiment, tips }

8. Safety, Quality & Guardrails

- Do not provide medical/legal advice. - Add confidence thresholds for AI suggestions. - Log and review hallucinations. - Provide manual override for mentors.

9. Evaluation Metrics

- RAG accuracy (grounded answers) - Booking success rate - Session completion rate - User satisfaction (CSAT) - Latency & cost per query

10. FAQs

Q: Can AI replace mentors? A: No. AI provides guidance when mentors are unavailable. Q: How to prevent wrong advice? A: Use RAG grounding, disclaimers, and mentor review loops.